Precision Control of Robotic Arm Using ESP8266

A Report submitted in the partial fulfilment of the requirement for

Real Time Project (B.TECH)

in

Electronics And Communications Engineering

Submitted by

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CERTIFICATE BY THE SUPERVISOR

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CONTENTS

ABSTRACT

This project aims to build a Wi-Fi-enabled robotic arm using the ESP8266 microcontroller. This design allows for wireless control through a web browser. The ESP8266 has built-in Wi-Fi and hosts a local web server. The arm consists of 3D-printed parts and is mounted on a perf board. It has 6 degrees of freedom (DOF): a rotating base, shoulder, elbow, rotator, wrist and a gripper. These movements are powered by SG90 and MG995 servo motors. Users can manage the arm through a straightforward HTML-based dashboard. This dashboard sends movement commands directly to the ESP8266. The microcontroller then processes these commands and adjusts the servo motors. This setup offers an affordable and compact way to control physical systems from a distance, which is perfect for small automation projects.

It also shows how well hardware and software can work together. In summary, this project emphasises the potential and versatility of IoT. It demonstrates how devices like the ESP8266 can add intelligence and connectivity to everyday machines. It also shows how well hardware and software can work together. In summary, this project emphasises the potential and versatility of IoT. It demonstrates how devices like the ESP8266 can add intelligence and connectivity to everyday machines.

LIST OF FIGURES

LIST OF TABLES

CHAPTER-1: INTRODUCTION

The purpose of this paper is to develop a model that relates desired performances to required components, their composition, and their cost, making it possible to design and produce high-quality robotic arms at the right cost.

1.1 BACKGROUND AND MOTIVATION

Imagine a world where there are no scientific inventions and the world is still primitive. People have to work tirelessly, and it takes a lot of time. So we need the necessary technology to make our lives simple. When these technologies are further refined, we can even create artificial machines that can imitate human motion and gestures. Inventions like batteries, wheels, and elements brought a new era in which everything is possible. In order to speed up things over the past 1000 years, humans made progress in a much drastic way. A few compelling ideas constitute a history of robotics:

These robots are constructed and designed in different shapes and sizes. One such design that can be used efficiently is the robotic arm, which resembles the human arm, and it can be used for performing human actions like grabbing and picking up things. Not only are these useful for industrial purposes, but also in medical purposes, like helping disabled people to regain their arms again.

In the end, robotics is not only the machines themselves. It concerns the within us and how we endeavour and wish to shape, create and mould the future. Transforming knowledge into tangible things, transforming theories into reality with imagination and effort.

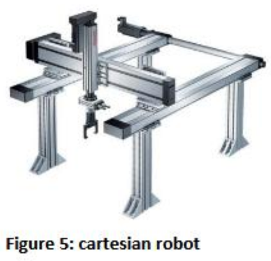


Figure . 1.1.1

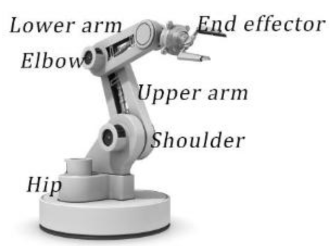


Figure . 1.1.2

1.2 THE ROBOTIC ARM TYPE:

The articulated robotic arm is a type of robotic arm which contains 4 Degrees of freedom it has. In total, there are 6 degrees of freedom, 3 translating in the x, y and z direction, and three rotations around these axes. This is the basic prototype that is generally used in the present time.

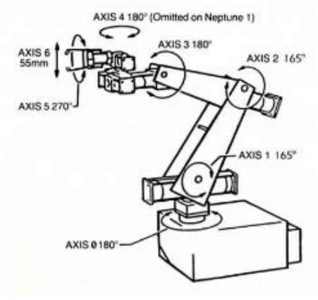


Figure 6: articulated robot

Figure . 1.2.1

From the figure above, we can see that the design itself is a bit complicated, but it gives a basic understanding of the working and other necessary details.

A robotic arm is fundamentally structured to replicate human arm movements, comprising several interconnected segments or "links" joined by various "joints" that enable rotational or linear motion, defining the arm's degrees of freedom.

The robotic arm consists of various components like motors, sensors, joints and the outer skeleton is designed to withstand the force and movements. It contains a total of 6 parts: The base, shoulder, elbow, rotator, wrist, and gripper/ End Effector.

The movements over angle are dependent on the motors used, and the control is given by web web-controlled interface, which is handled from the user's side.

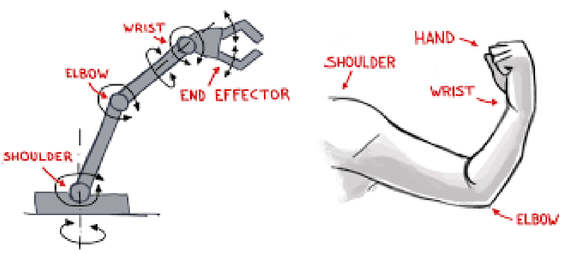


Figure . 1.2.2

CHAPTER-2: LITERATURE REVIEW

2.1 Overview

Controlling robotic arms through the internet is becoming more popular, especially in education and DIY projects. With the help of Wi-Fi-enabled microcontrollers like the ESP8266, it's now easier and cheaper to build robotic systems that can be operated remotely. In this literature review, we look at how these systems have been built in the past, what their limitations are, and how our project offers improvements.

2.2 Existing Systems

Many low-cost robotic arms use Arduino Uno, Raspberry Pi, or similar microcontrollers with servo motors to perform basic movements. Most systems are limited to 2 or 3 degrees of freedom (DOF) and are controlled through wired connections, Bluetooth modules, or basic mobile apps. While Bluetooth control is wireless, it has a short range and requires manual pairing, which limits flexibility and scalability.

User interfaces in these systems are usually basic desktop apps or mobile applications and rarely include web-based control. Most do not provide features such as motion recording or automated playback, and many operate without position feedback or modular expandability.

2.3 Comparison with Our Project

Our project uses the ESP8266 microcontroller, which functions both as the controller and as a Wi-Fi access point. Instead of connecting to an external router, the ESP8266 creates its own Wi-Fi hotspot, allowing users to connect directly using any device with a browser. This enables wireless control without any internet connection or additional network setup.

The control interface is a webpage hosted by the ESP8266, featuring sliders for manual control and buttons such as Start Recording, Stop Recording, and Play. These buttons allow users to record a series of movements and replay them automatically, simulating basic automation.

Servo control is handled through an external PCA9685 servo driver, which ensures smooth and reliable PWM signals while offloading work from the ESP8266. This allows the use of multiple servos without signal conflict or overload.

Compared to earlier systems, our project includes:

1) Web interface served by ESP8266's Wi-Fi hotspot
2) No need for apps, pairing, or external networks
3) Playback system for recorded movements
4) Improved stability using PCA9685 driver
5) Support for more joints with future expandability

These upgrades make the system more flexible, interactive, and scalable than traditional Bluetooth- or USB-controlled robotic arms.

CHAPTER 3: SYSTEM OVERVIEW

The robotic arm system is based on the Internet of Things (IoT) concept, enabling wireless control of servo motors through a web interface hosted on the ESP8266 microcontroller. The system is designed to provide a low-cost, browser-based control solution for a 6-DOF robotic arm using HTML, CSS, and JavaScript to build the user interface and PCA9685 to generate precise PWM signals for servo motors.

The ESP8266 acts as both the server and the controller. It hosts a local Wi-Fi network to which any browser-enabled device (such as smartphones or laptops) can connect. No external router or internet connection is required, making it suitable for standalone and offline use. The system supports not only manual control but also recording and replaying motion sequences using "Start Recording," "Stop Recording," and "Play" buttons on the interface, which enhances its functionality for automation and demonstrations.

3.1 BLOCK DIAGRAM(SYSTEM FLOW)

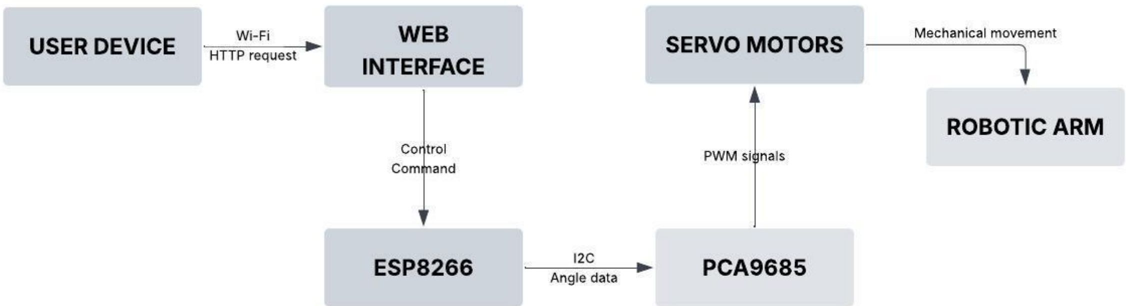


Figure 3.1

The ESP8266 microcontroller is powered by a 5V micro-USB power supply. It also acts as the central controller for the robotic arm’s movements and oversees both the network communication and the microcontroller's functions.

With power on, the ESP8266 either connects to or creates a wireless network, which allows communication between the robotic system and user devices: smartphone, tablet, or computer. The user can initiate a web interface that the ESP8266 hosts, using any device with a browser.

From the web interface, the robotic arm is under user control by sending direction control commands. The commands are processed by the ESP8266 and appropriate PWM signals are produced.

The generated PWM signals are sent to six servo motors which control joint movements of the robotic arm:

Servo 1: Controls base rotation

Servo 2: Controls shoulder joint

Servo 3: Controls elbow joint

Servo 4: Controls wrist pitch

Servo 5: Controls wrist roll

Servo 6: Controls the gripper mechanism

These servos make up the movement system of a 3D-printed robotic arm, permitting complex articulated movements that can be controlled wirelessly by a user.\

3.2 SYSTEM ARCHITECTURE

The system follows a client-server architecture:

● Client: A mobile phone or laptop that interacts with the system using an HTML interface on a web browser.

● Server: The ESP8266, which serves the webpage, hosts and receives user inputs via HTTP GET requests.

● Upon receiving the requests, the ESP8266 does the necessary processing and generates the PWM signals needed to drive the individual servo motors. Operating under this architecture provides the capability for quick responding controls, independence from cloud infrastructures, and enhanced offline durability.

3.3 HARDWARE-SOFTWARE INTERACTION

The components of a computer system, with hardware and software interfaces, interact as follows:

● The web interface of the system is implemented in CSS, HTML, and JavaScript, which are part of the ESP8266 firmware.

● Directional buttons on the webpage generate HTTP requests, which are sent to the ESP8266 controller.

● The ESP8266 firmware, coded in C++, processes these requests and sends proper commands corresponding to predefined servo angles.

● The commands are carried out by the hardware devices (ESP8266 and servo motors) through the robotic arm's motion controlled by the PWM signals. This describes the complete flow of interaction between the user and the robotic arm control system. This method makes sure that the robotic arm is accurately controlled in real-time with respect to user actions done through the interface.

CHAPTER 4: IMPLEMENTATION

4.1 DESIGN PROCEDURE

4.1.1 SYSTEM STUDY AND ANALYSIS

The reference video was thoroughly analyzed to understand the structure, motion, and components of the robotic arm, including the base, joints, links, and end-effector.

4.1.2 DESIGNING OF ROBOTIC ARM

The robotic arm model consists of many individual sections/parts which are combined together to form the overall model. The parts of the model are first simulated and then these components are 3D printed and assembled together.

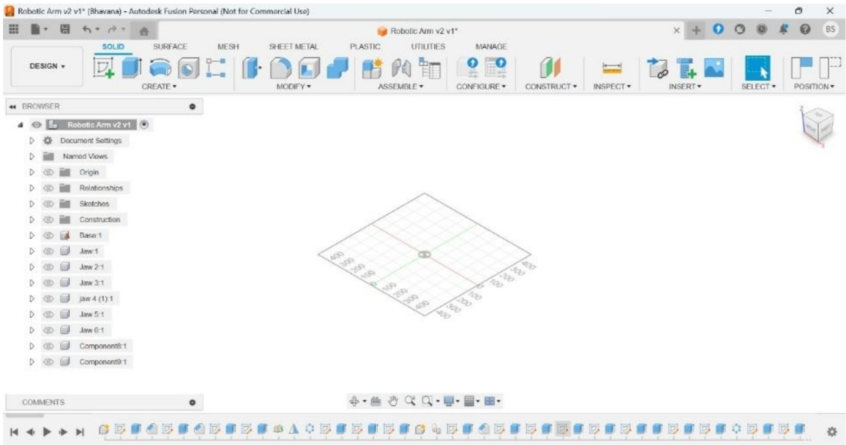


Figure 4.1.1 (Fusion 360)

Step-1:

Let us understand the process of designing the arm. First we will design the 2D models in FUSION 360 Software. The 2D shapes are drawn using a variety of commands, such as line, rectangle, arc, etc and are then extended to required sizes. The components like the Base , Link 1, Link 2, Rotator, Joint, Gripper are designed likewise.

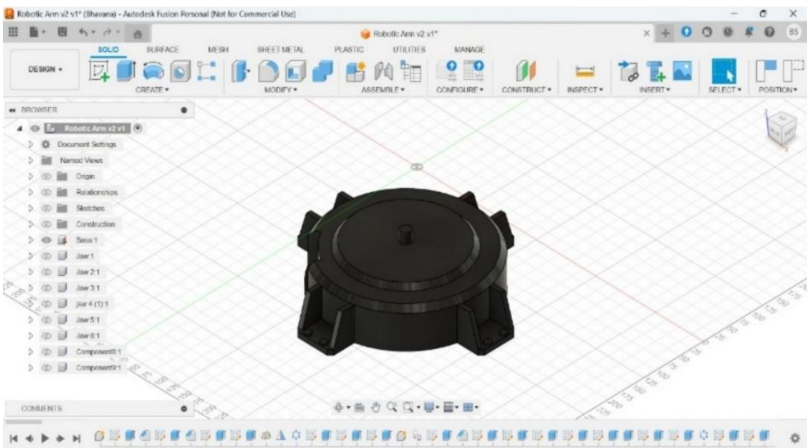


Figure 4.1.2 (Base of the arm)

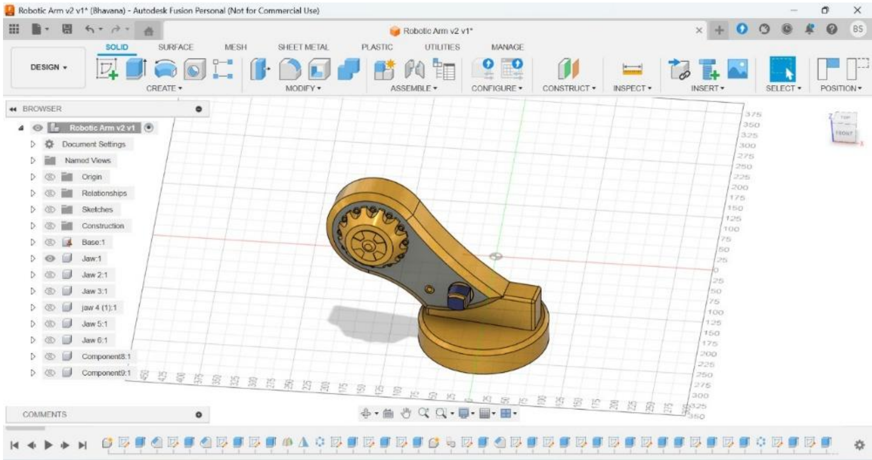


Figure 4.1.3 (Link 1)

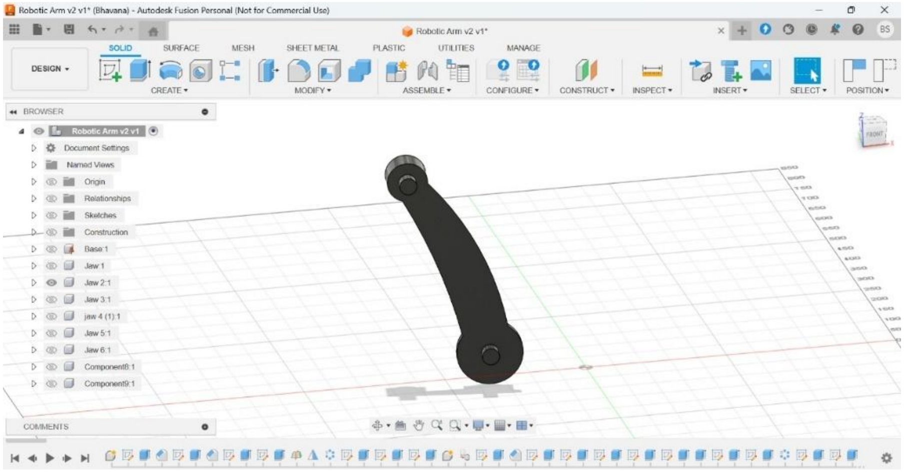


Figure 4.1.4 (Link 2)

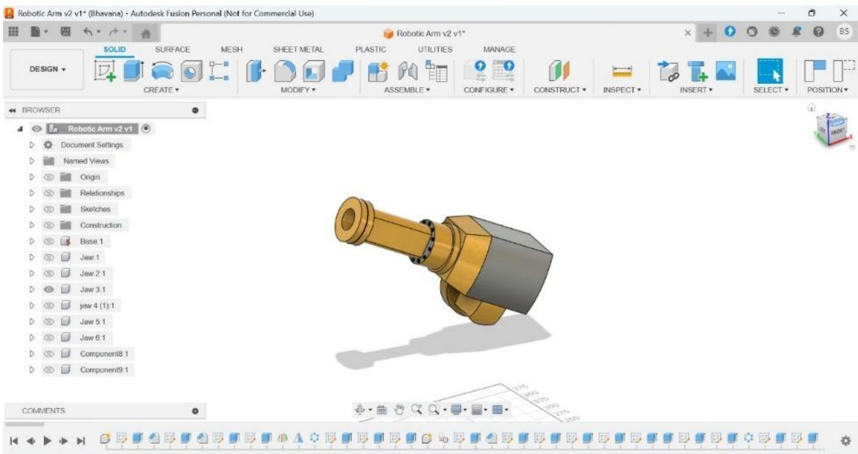


Figure 4.1.5 (Rotator)

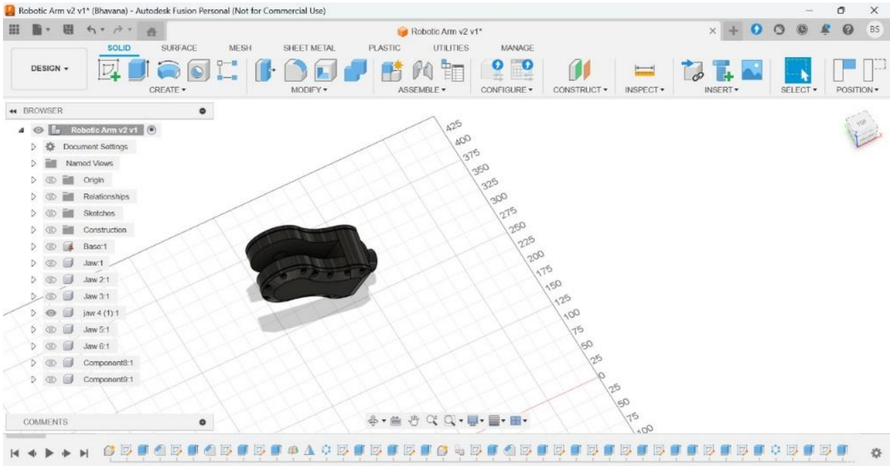


Figure 4.1.6 (Joint)

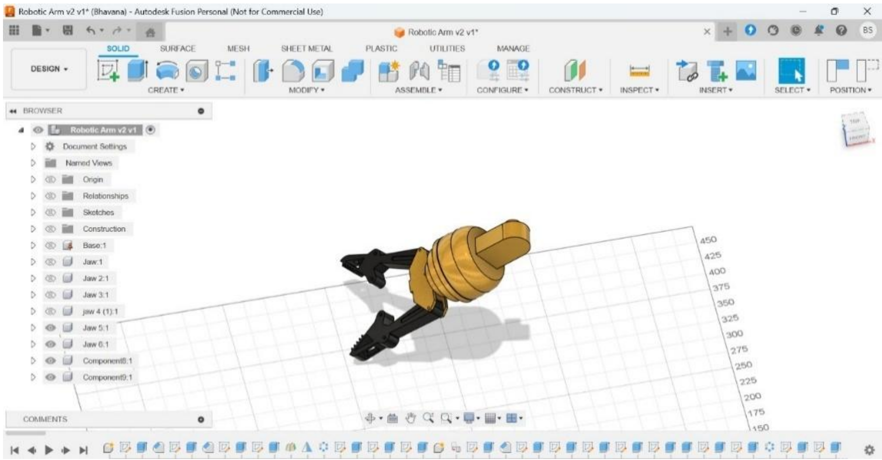


Figure 4.1.7 (Gripper)

These 3D printed components are then combined together to form the overall 3D printed arm model which is shown in the figure 4.2.8.

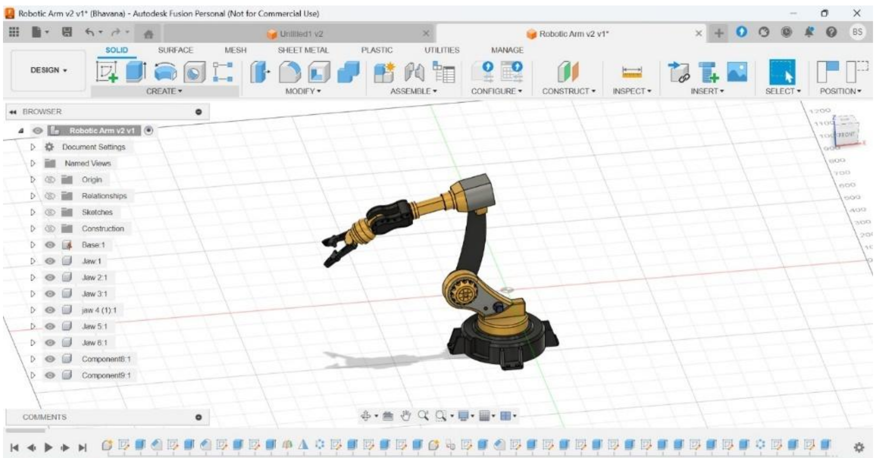


Figure 4.1.8

This model is then combined with the rest of the hardware components like ESP8266, Servo Motor Driver, Motors(SG90, MG995) with the help of jumper cables.

4.2 HARDWARE COMPONENTS:

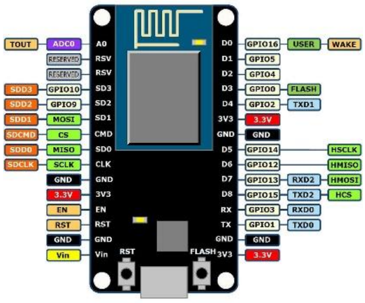
<table>
<tr><th colspan="2">HARDWARE COMPONENTS</th><th>DESCRIPTION</th></tr>
<tr><td>1) ESP8266</td><td>


Figure 4.2.1</td><td>The ESP8266 is a low-cost and low-power-consuming System on Chip microcontroller which is integrated with wi-fi connectivity.</td></tr>
<tr><td>2)MG995 MOTOR</td><td>

Figure 4.2.2</td><td>Basically, this is a high-torque, metal geared servo motor which is used in heavier load applications like robotic arms</td></tr>
<tr><td>3)MG MOTOR</td><td>

Figure 4.2.3</td><td>Compared with MG995, this servo offers very low torque, so it is used in low-load applications.</td></tr>
<tr><td>4)SERVO MOTOR DRIVER</td><td>

Figure 4.2.4</td><td>This is a servo motor driver (PCA9685), which consists of 16channels for controlling of the servo motors.</td></tr>
<tr><td>5)3D PRINTED STRUCTURE</td><td>

Figure 4.2.5</td><td>The outer skeleton which consist of the different individually designed sections.</td></tr>
</table>

Table 4.2 : Hardware components

4.3 CIRCUIT CONNECTIONS OF ROBOTIC ARM

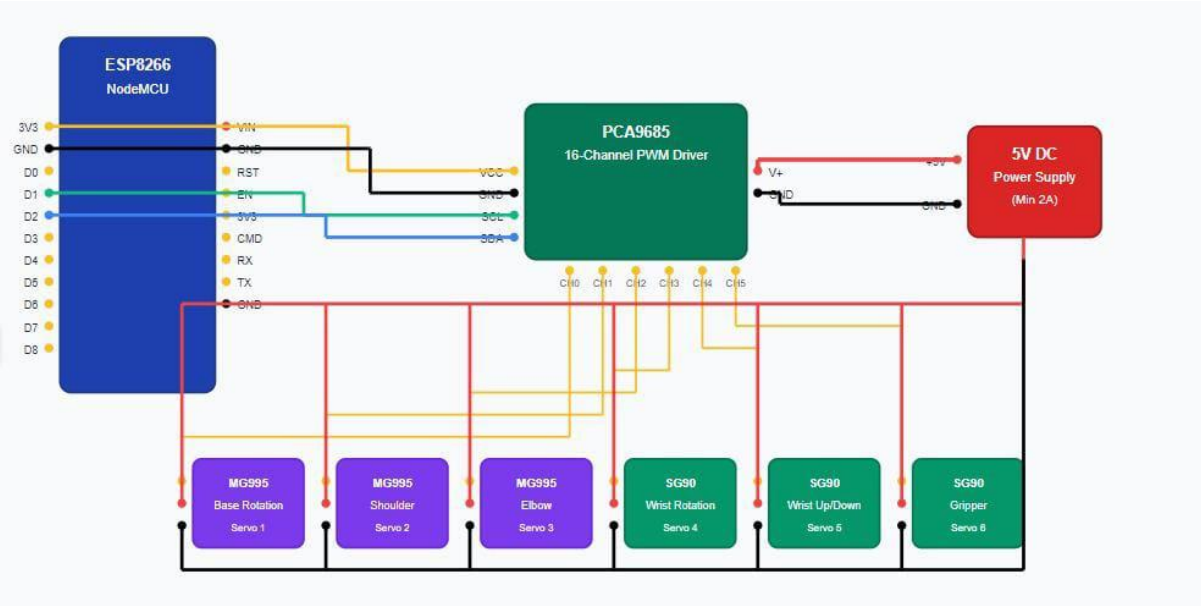


Figure 4.3

The circuit connections are shown in the figure 4.3.

- The Vcc and ground pins of the ESP8266 and Servo Motor Driver are short circuited and the value of +5V and 0V respectively.
- The Servo Motor Driver(PCA9685) consists of [SCL and SDA] pins and [V+ and GND] pins. The input to [SCL and SDA] pins are given by the digital pins (D0 ,D1) from the ESP8266.
- The input to [V+ and GND] pins are given by the [+5v and GND] of the power supply.
- Any 6 pins from the 16 pins of the Servo Motor Driver(PCA9685) are connected to each of the 6 motors.

CHAPTER 5: WORKING PRINCIPLE

The working principle of our Web-Controlled Robotic Arm revolves around using the ESP8266 microcontroller as both a controller and a wireless web server, along with the PCA9685 servo driver to precisely control multiple servo motors.

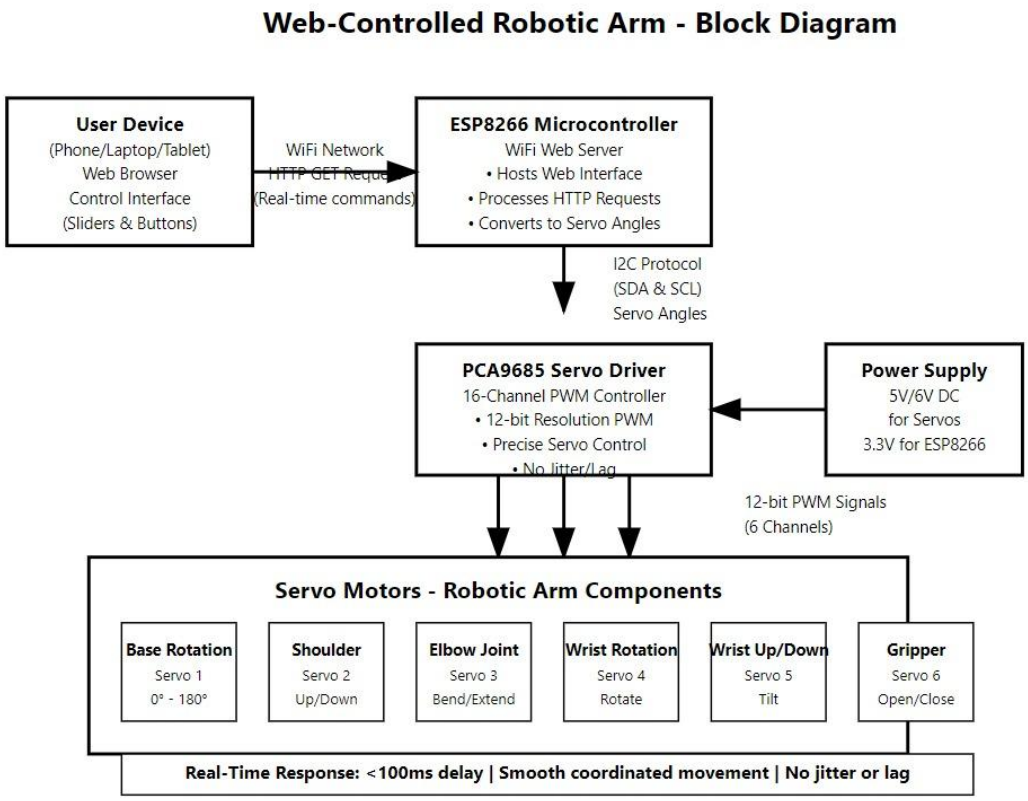


Figure 5.1

5.1 CONTROL FLOW

The ESP8266 first sets up a local Wi-Fi hotspot or connects to an existing Wi-Fi network. It hosts a responsive web interface that can be accessed from any device with a browser (like a phone or laptop). This webpage includes sliders and buttons for controlling each joint of the robotic arm.

When a user interacts with these sliders, the browser sends HTTP GET requests containing angle values to the ESP8266. These requests are received in real-time and processed immediately.

5.2 SERVO ACTUATION LOGIC

Instead of generating PWM signals directly from the ESP8266 (which has limited and less stable PWM capability), the microcontroller sends the processed servo angles to the PCA9685 driver module via the I2C communication protocol.

The PCA9685 then generates high-resolution 12-bit PWM signals to control each servo motor. Each motor is assigned to control a specific part of the robotic arm:

1) Base rotation
2) Shoulder (up/down movement)
3) Elbow joint
4) Wrist rotation
5) Wrist up/down
6) Gripper open/close

This separation of tasks improves stability, accuracy, and performance.

7.3 REAL-TIME COMMAND EXECUTION

Due to the efficiency of I2C communication and dedicated PWM control by PCA9685, the system responds in real-time. The delay between moving a slider on the webpage and actual servo movement is typically less than 100 milliseconds, which provides a smooth and responsive user experience.

Even if multiple servos are moved at the same time, the PCA9685 handles them without any lag or jitter, allowing precise and coordinated movement of the robotic arm.

CHAPTER-6:

RESULTS AND DISCUSSIONS

The Web-Controlled Robotic Arm System using ESP8266 was tested and gave successful results. The main goal was to move the robotic arm wirelessly using a web-based interface, and this goal was achieved. Users could control the robotic arm through a simple web page using sliders and buttons, without needing any physical connection. The interface worked on different devices like smartphones, laptops, and tablets.

Each slider controlled one servo motor: one for the base rotation, one for the elbow joint, and one for the gripper. When a slider was moved, the ESP8266 received the command and sent the correct PWM signals to the servos. These signals made the motors move to the desired position.

The robotic arm responded correctly and smoothly to the commands most of the time. The movement of the servos matched the input from the user without noticeable delay. The system worked reliably and was easy to use, showing that it is a good option for wireless control of a robotic arm using a web browser.

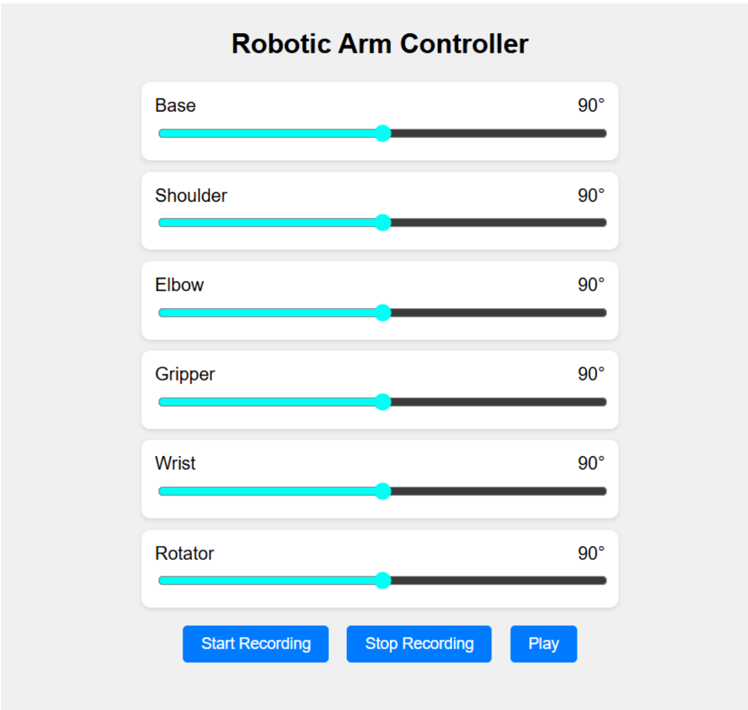


Figure . 6.1

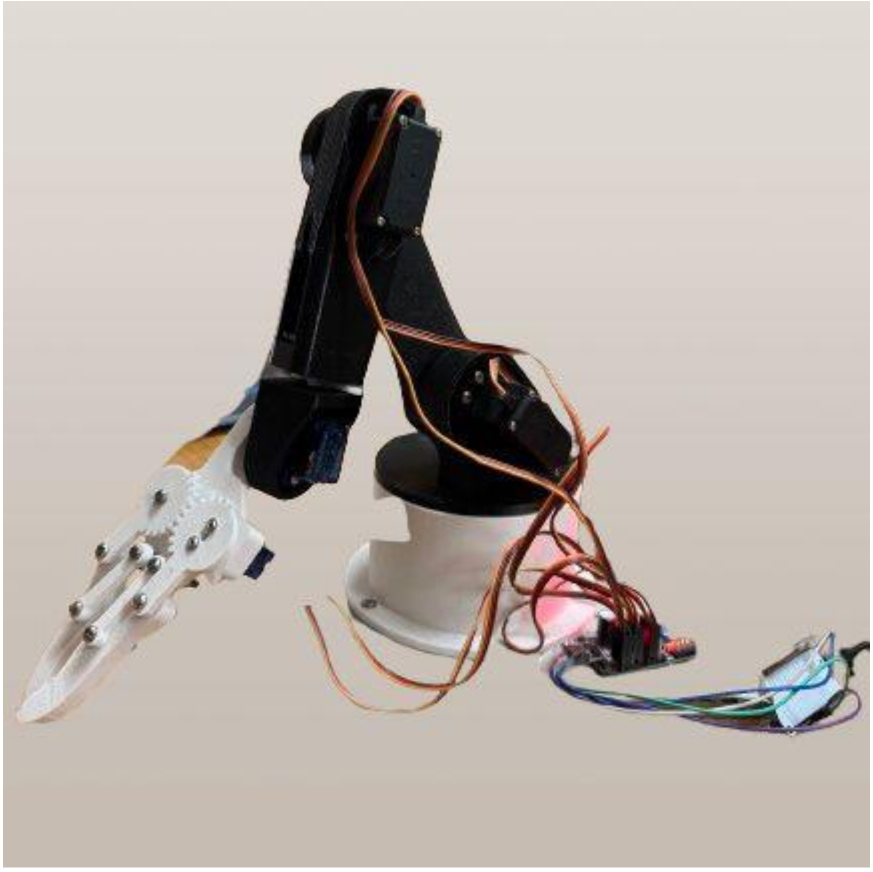


Figure . 6.2

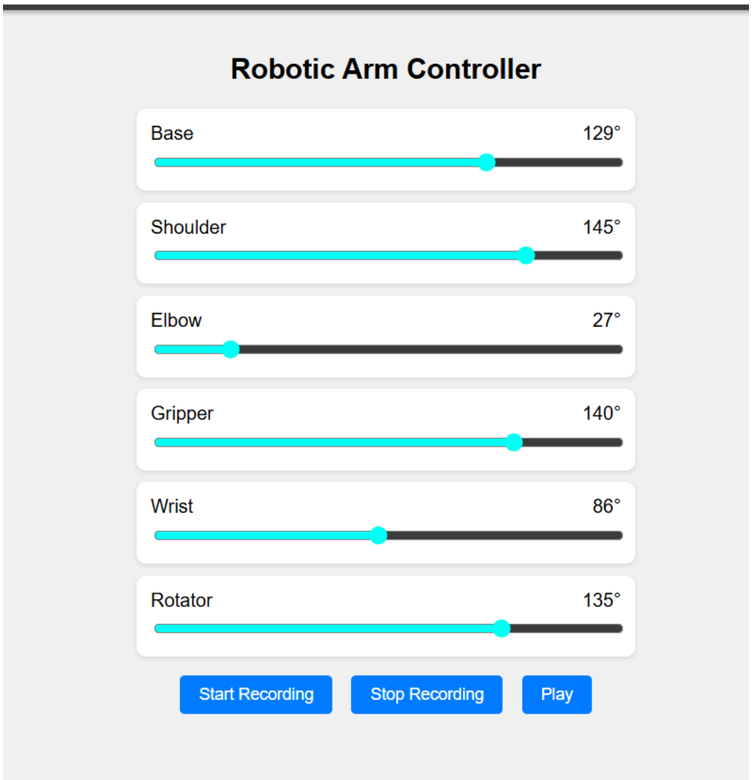


Figure 6.3

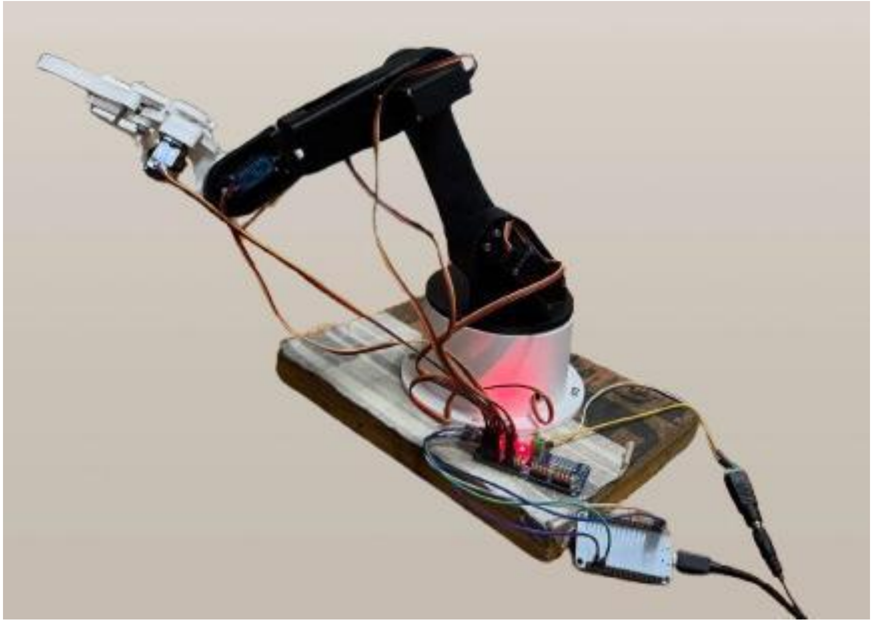


Figure . 6.4

Readings and Observations:

The tables below contains the readings which are noted by observing the movements and positions of the motors .

Parameters	Measured values
WIFI connection	13 sec
Webpage loading	1.5 sec

Table 6.1 : Measured values of parameters

Joints	Input Angle	Output Angle	Error	Accuracy
Elbow	30°	25°	5° error	83.34%
Base	45°	43°	2° error	95.56%
Rotator	45°	42°	3° error	93.4%
Gripper	24°	19°	5° error	79%
Wrist	30°	26°	4° error	86.7%

Table 6.2: Accuracy calculations

CHAPTER-7:
CONCLUSION AND FUTURE SCOPE

The Web Controlled Robotic Arm using ESP8266 has been developed to enable remote control of mechanical operations through a user-friendly web interface. By utilizing the Onboard Wi-Fi capability of the ESP8266 microcontroller, the system allows real-time control of a robotic arm without the need for complex hardware or direct human intervention contact. The wireless control mechanism not only enhances flexibility but also ensures safer handling in conditions where human Presence may be risky or impractical.

The system is cost-effective, easy to deploy, and requires minimal technical expertise to operate. It offers a scalable solution that can be further enhanced with features such as visual feedback using a camera, integration of sensors for precise control, and cloud or mobile-based remote access.

In conclusion, this project successfully demonstrates the practical application of remotely operated systems in automation. It serves as a foundation for future developments and reflects the growing importance of accessible and intelligent control solutions in modern technology.

7.1 Key Outcomes of the Project:

1)Successfully designed and implemented a web-based robotic arm system using ESP8266.

2)Enabled remote control of robotic arm joints (base, elbow, gripper) through a responsive website.

3)Integrated MG995 and SG90 servo motors for better torque management and precision.

4)Achieved real-time response between user input and motor action using HTTP requests.

5)Eliminated the need for mobile apps or software installations—works with any browser.

6)Ensured modular code design to allow future expansion of joints or sensor feedback.

7)Developed a functional prototype suitable for academic learning and IoT-based automation projects.

7.2 Future Scope:

Web Interface Enhancements
Add interactive 3D simulation, joystick controls, or live angle feedback to improve the user experience.

Voice Command Control
Integrate voice recognition modules or browser-based voice input for hands-free operation.

Mobile App Development
Create a dedicated Android/iOS app with advanced features like saving positions or gesture control.

Cloud Connectivity
Add MQTT or WebSocket support for cloud-based monitoring and control from any global location.

REFERENCES

[1] Jadhav, A., & Patil, S. (2021).” IoT-Based Robotic Arm Controlled Through Web Interface”. International Journal of Innovative Science and Research Technology (IJISRT), 6(4), 233–236.DOI: 10.5281/zenodo.4737857

[2] Sahu, S., & Dash, M. K. (2019).” Web-Based Control of Robotic Arm Using IoT for Remote Applications”. Proceedings of the 2019 IEEE International Conference on Intelligent Systems and Control (ISCO), 1–5.DOI: 10.1109/ISCO.2019.8716678

[3] https://github.com/rajtilak-2020/Robotic_Arm .

[4] An enquiry into the relation between components and their composition on the desired performance and cost of pick-and-place robotic arms. Teun Schurink,10-2016,